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Department of Computer Engineering
ARCHITECTURE AND ORGANIZATION OF COMPUTER SYSTEMS
WORKSHEET - 3

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Q1: What is ALU? Discuss in detail

The Arithmetic Logic Unit (ALU) is a part of the CPU that performs arithmetic and logical operations on data. It takes input values, performs the selected operation, and produces a result.

- **Arithmetic operations:** addition, subtraction and multiplication.
- **Logical operations:** AND, OR and NOT. The ALU can also compare values and shift bits.

The ALU contains smaller units, such as an adder and logic gates. An operation code selects which unit's result appears at the ALU output. For example, one code selects addition, while another selects AND.

Q2: How do we draw K-maps, and what is their purpose in computer architecture?

A Karnaugh map (K-map) helps us simplify a Boolean expression. A simpler expression can use fewer logic gates when we build a circuit, such as an adder.

First, make a truth table for the inputs and outputs. Draw one K-map for each output. Label its rows and columns in Gray-code order (00, 01, 11, 10), and fill in the output values. Group neighboring 1s in groups of 1, 2, 4 or 8. Then write the simplified expression.

For example, a 2-bit adder has four input bits and three output bits. We draw three 4×4 K-maps, one for each output bit.

Q3: What is the Bit-Slice Approach?

The bit-slice approach builds a large circuit from many copies of a small 1-bit circuit. For an adder, we design one 1-bit full adder and use one copy for each bit position.

Each full adder takes two input bits and a carry-in. It produces one sum bit and a carry-out. The carry-out goes to the next slice. For example, a 32-bit adder uses 32 one-bit slices. This is easier than designing one huge circuit with K-maps.

Q4: What is the difference between a Ripple Carry Adder and a Carry Look-Ahead Adder?

- **Ripple Carry Adder (RCA):** the carry passes through each 1-bit adder in order. It is simple, but it becomes slow when there are many bits.
- **Carry Look-Ahead Adder (CLA):** it uses generate and propagate signals to calculate carries earlier. It is faster, but it needs more gates and is more complex.

Both add binary numbers. The main difference is how they calculate the carry between bit positions.

Q5: What do carry propagation and carry generation mean?

Carry generation means a bit position creates a carry on its own. For example, $1 + 1$ produces a carry of 1, even when no carry comes in from the previous position.

Carry propagation means a bit position passes an incoming carry to the next position. For example, when its input bits are 1 and 0, a carry from the previous position continues to the next one.

When both input bits are 0, the position does not create or pass a carry. Carry look-ahead adders use these two ideas to work out carries faster.